



WAYANAMC EDUCATION TRUST®
DON BOSCO INSTITUTE OF TECHNOLOGY
Kumbalagodu, Bengaluru – 560 074



Department of Electrical and Electronics Engineering

Project Presentation on

**Solar-Powered Autonomous Grass Cutter with Obstacle
Detection and Real-Time Monitoring**

Solar-Powered Autonomous Grass Cutter with Obstacle Detection and Real-Time Monitoring

Team members

1DB21EE014 – KISHOR C

1DB21EE027 – SRIKANTH S

1DB22EE415 – SAGAR S A

1DB22EE419 – VENUGOPALAREDDY G S

Guided By

Dr. R ANGURAJA

HOD, Dept. of EEE

DBIT, BENGALURU

Prof. R SANTHOSH KUMAR

Assistant Professor, Dept. of EEE

DBIT, BENGALURU

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ABSTRACT

The Solar-Powered Autonomous Grass Cutter with Obstacle Detection and Real-Time Monitoring robot designed to efficiently maintain lawns and fields. Powered by a polycrystalline solar panel and a backup battery, it uses ultrasonic sensors for obstacle detection and avoidance, ensuring safe operation. The grass-cutting mechanism, driven by a high-speed BLDC motor, is effective and energy-efficient. An LCD display provides real-time updates on the robot's condition and obstacle status, enhancing user monitoring and control. This project combines renewable energy with automation to reduce human effort and promote sustainable practices. Designed to prevent damage to objects, humans, or animals. System integrates Arduino ATmega328p, ultrasonic sensors, and motor driver circuit for smooth operation.

INTRODUCTION

- The grasses are mainly grown in parks, colleges, fields for aesthetics, safety, hygiene and appearance.
- The grass grows very fast and it has to be maintained properly, insects and snakes will increase and hence it has to be cut frequently.
- Nowadays the world is facing labor shortage problem for manual workers.
- To address this problem we need an autonomous system which works on its own to cut the grass without human interference.
- It utilizes solar energy to power both the robot and the cutting mechanism, making it an eco-friendly alternative to fuel-powered mowers.
- The system incorporates ultrasonic sensors for obstacle detection, ensuring safe operation.
- By using renewable solar energy, it reduces operational costs and environmental impact.
- High-speed BLDC motor for fast and effective grass cutting and relay control for precise on/off operation of the cutting mechanism.

LITERATURE SURVEY

Sl. No	Author , Title, Publication Year	Highlights	Limitations
1.	Lingappa j, V. Raghavender, G Aparna and Ch. Manideep Automated grass cutter using renewable energy, May-2021	This paper showcases the development of automatic grass cutter. The system is powered by a 12V 7.5Ah lead-acid battery, rechargeable through solar energy harvested by a connected 12V, 10W solar panel. Noteworthy for its cost effectiveness, robust design, and durability, the system offers a reliable and efficient solution for grass-cutting needs.	• It does not have any safety measure if any obstacle occurs in front of grass cutter.
2.	Rakesh poojari, Rajesh kumar, Hitesh B.patil, Fully Automated Solar Grass Cutter, Nov-2020	In this model, source is driven from the solar energy using photo voltaic panel which charges the battery and is utilized for powering operation of the system. The system controlled by P89V51RD2 micro controller. L298 driver circuit is used for compatibility of micro controller and motor. Toggle switch for movement operation .DPDT switch for movement operation. Wheel chain with 26 links on both sides are attached to platform support the whole model. In wheel two motors are used with 45 rpm. By this automated solar cutter is fully automated grass cutting robotic vehicle powered by solar energy.	• It is complex in design. • It is expensive. • It has slower movements and less efficiency

Sl. No	Author , Title, Publication Year	Highlights	Limitations
3.	Mallikarjun, Mudda, Vishwateja, Srujankumar, Praveen kumar , Automated Solar Grass Cutter, Jun-2020	In this model, 10 watts solar panel is used to charge the batteries which are rechargeable , microcontroller 8051 and motor driver. .The motor are driven by using motor driver. It is also known as H-bridge .The main purpose of using motor driver is because that DC motors require the minimum voltage as 9v as input .But the micro controller gives output as only 5v so we require 9v to 12v for driving the motors .So we use motor driver which takes 5v as input and gives the 12v for motors .The L293D motor driver drives only two motors which can move in both direction .And the cutting blade is used to cut any type of grass .they used 1400rpm motor for cutting blade .The motors runs directly by 12v rechargeable battery.	• It does not have any safety measure if any obstacle occurs in front of grass cutter. • It is low efficiency
4.	Ayesha Sultana, Shireen Fatima, Hajara Mubeen, Smart IoT based Gesture Controlled Grass Cutting Vehicle ,Dec-2021	This paper explains the solar grass cutter using hand gesture, solar powered robotic grass cutting machine that eliminates the obstacles and could be accessed from every direction by utilizing hand gesture through remote camera which is introduced to the highest point of the motor-vehicle to communicate remote gushing to the users end. The system uses a 6v batteries to drive both the motor vehicle movement and the grass cutter engine.	• It requires a human intervention • It is expensive

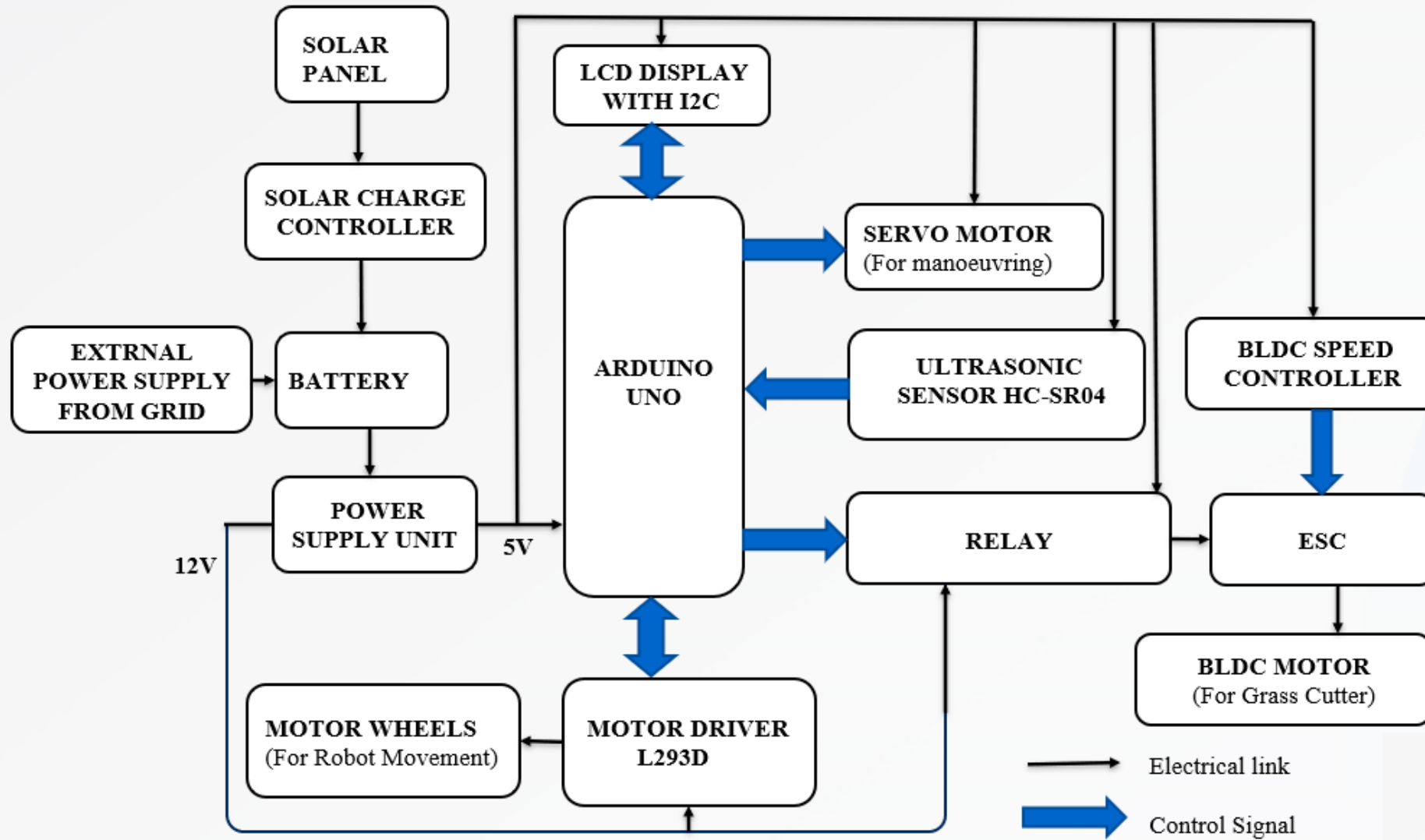
PROBLEM STATEMENT

To overcome the problems faced due to shortage of labours, a new approach is required. The current reliance on fuel-powered lawn mowers is a significant concern. These machines contribute to environmental pollution, high operational costs, and noise pollution. As fuel prices rise and emissions increase, an alternative solution is necessary. This project proposes a solar-powered grass cutting machine that eliminates fuel needs

OBJECTIVES

- To overcome the problems faced due to shortage of labours.
- Develop an autonomous grass cutting system with less human intervention.
- Use renewable solar energy, promoting sustainability in everyday tasks.
- Enhance cutting efficiency with high-speed, for faster grass trimming.
- Integrate safety features that stop the blade motor when obstacles are detected.

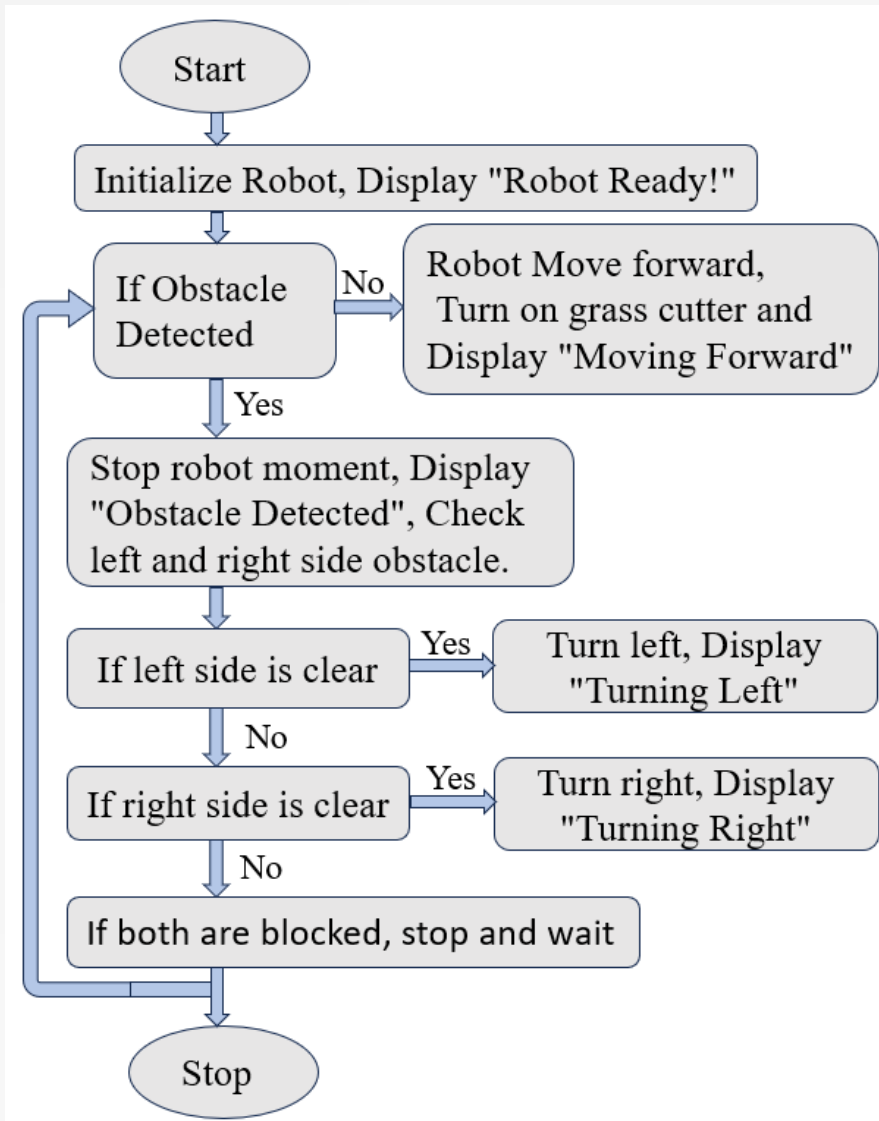
BLOCK DIAGRAM



METHODOLOGY

- **Power Supply:**
 - Solar panel charges the battery via the charge controller.
 - 12V battery powers motors, while 5V powers the Arduino, sensors etc.
- **Arduino Uno:**
 - Central controller for motor movements, sensors, LCD display.
- **LCD Display (16x2 with I2C):**
 - Display the obstacle condition and robot status.
- **Motor Driver (L293D):**
 - Controls left and right DC motors for movement (forward, left, right).
 - EN1 and EN2 control motor speed (PWM).
- **Ultrasonic Sensor with Servo:**
 - Detects obstacles and rotates via the servo motor for 180° scanning.
- **Grass Cutter (BLDC Motor):**
 - BLDC motor, controlled via ESC, cuts grass when activated.
- **Workflow:**
 - Arduino reads inputs, avoids obstacles, and follows commands to move the robot.
 - Robot status are updated on the LCD.

Flow Chart:-



➤ Initialization

- Initialize Arduino, sensors, and LCD display.
- Display "Robot Ready!"

➤ Motion Control

- Move forward (activate DC motors).
- Turn on grass cutter motor (relay ON).
- Display "Moving Forward."

➤ Obstacle Handling

- Stop motors.
- Display "Obstacle Detected."
- Servo Motor scans left and right.

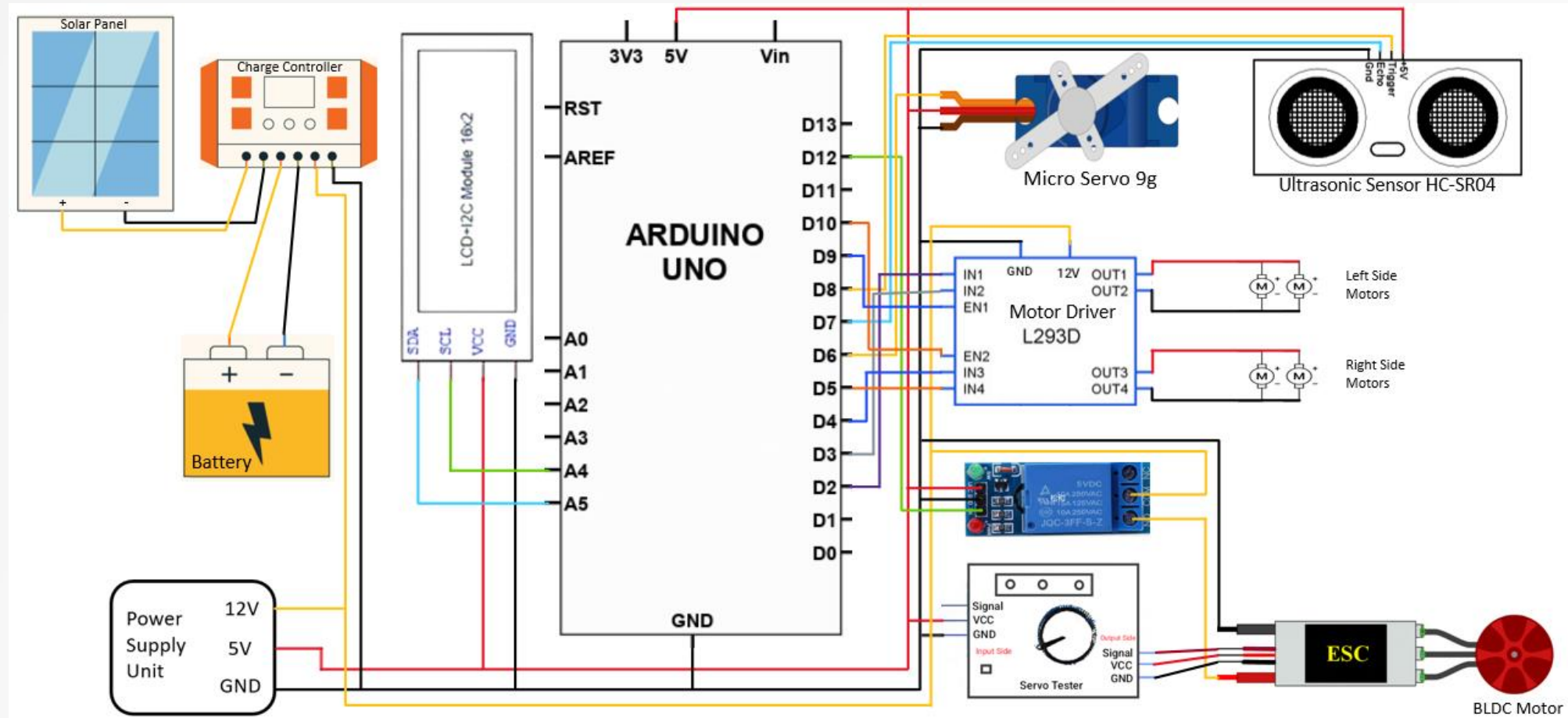
➤ Decision Logic

- If left is clear, turn left.
- If right is clear, turn right.
- If both are blocked, stop and wait.

➤ **Loop:-** Repeat Obstacle Detection and Decision Logic until power is insufficient or system is stopped.

➤ **Stop:-** Stop all operations.

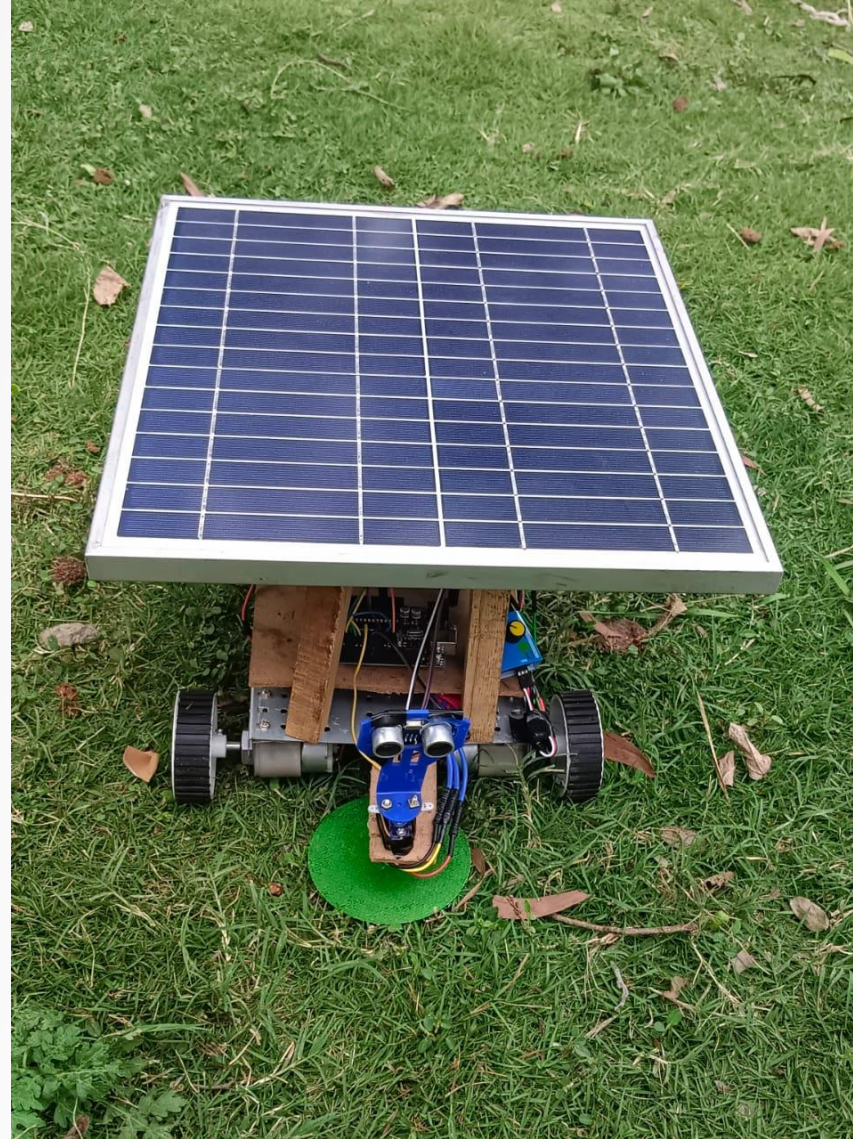
CIRCUIT DIAGRAM



RESULT



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OUTCOME OF PROJECT

- Labor Shortage Problem Solved: The project overcomes the problems faced due to shortage of labours.
- Autonomous Grass Cutting System: An autonomous grass cutting system is developed with less human intervention.
- Renewable Energy Utilization: Renewable solar energy is used, promoting sustainability in everyday tasks.
- Enhanced Cutting Efficiency: Cutting efficiency is enhanced with high-speed technology for faster grass trimming.
- Improved Safety: Safety features are integrated that stop the blade motor when obstacles are detected.

APPLICATIONS

- **Public Parks:** Efficient for maintaining large public green spaces without fuel costs.
- **Sports Fields:** Automates the upkeep of sports field lawns, reducing manual labor.
- **Garden Maintenance:** Suitable for personal or professional garden maintenance.
- **Commercial Landscaping:** Ideal for landscaping companies to automate grass cutting tasks.
- **Corporate Campuses:** Can be used on corporate campuses to reduce lawn maintenance costs.
- **Private Farms:** Used to maintain small to medium farm fields with minimal environmental impact.
- **Greenhouses:** Provides efficient and sustainable grass trimming in greenhouse environments.

CONCLUSION

- In conclusion, The Solar-Powered Autonomous Grass Cutter with Obstacle Detection and Real-Time Monitoring robot is an innovative solution that efficiently maintains lawns and fields, powered by a polycrystalline solar panel and a backup battery.
- The robot uses ultrasonic sensors for obstacle detection and avoidance, ensuring safe operation, and features a grass-cutting mechanism driven by a high-speed BLDC motor, which is effective and energy-efficient.
- This project combines renewable energy with automation to reduce human effort and promote sustainable practices, designed to prevent damage to objects, humans, or animals. Overall, the Solar-Powered Autonomous Grass Cutter represents a significant leap forward in lawn maintenance technology.
- This Project reduce noise pollution, and operational costs compred to desal lawn mover.

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PAPER PUBLICATION

- Title: Solar-Powered Autonomous Grass Cutter with Obstacle Detection and Real-Time Monitoring
- Abstract: The Solar-Powered Autonomous Grass Cutter with Obstacle Detection and Real-Time Monitoring robot designed to efficiently maintain lawns and fields. Powered by a polycrystalline solar panel and a backup battery, it uses ultrasonic sensors for obstacle detection and avoidance, ensuring safe operation. The grass-cutting mechanism, driven by a high-speed BLDC motor, is effective and energy-efficient. An LCD display provides real-time updates on the robot's condition and obstacle status, enhancing user monitoring and control. This project combines renewable energy with automation to reduce human effort and promote sustainable practices. Designed to prevent damage to objects, humans, or animals. System integrates Arduino ATmega328p, ultrasonic sensors, and motor driver circuit for smooth operation.
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 - ii. Guo-shing Huang und Keng-Chih Lin proposed “Intelligent auto saving energy robotic lawn mower & quote; IEEE transaction on robotics. Pg. 4130 to 4136 In 2020.

THANK YOU